Dynamical behaviour of Brownian particles coupled to a critical field

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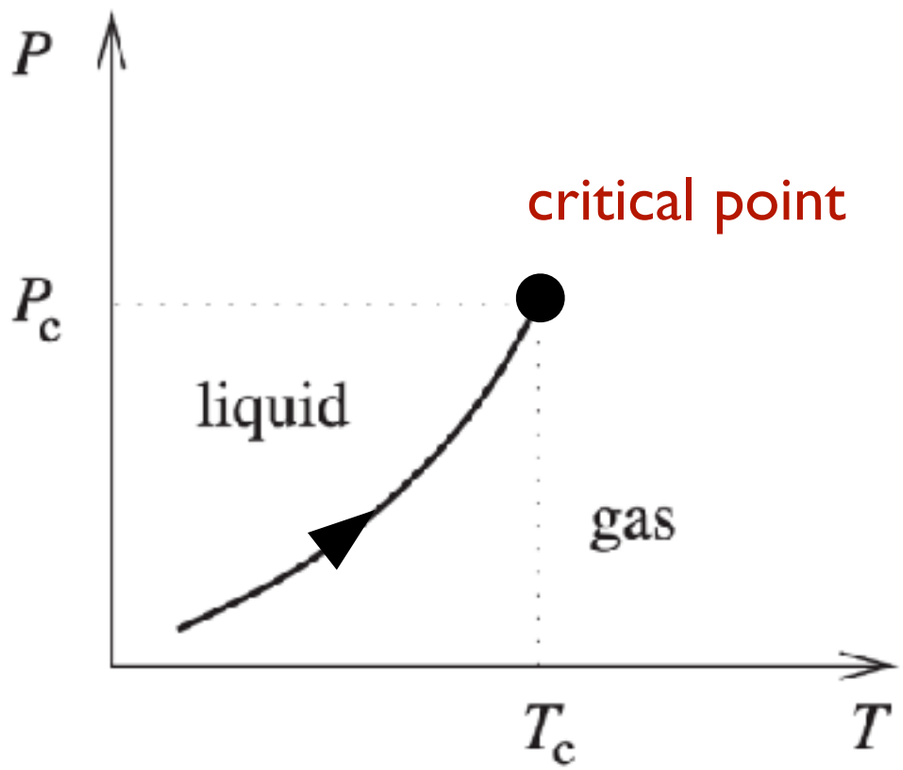
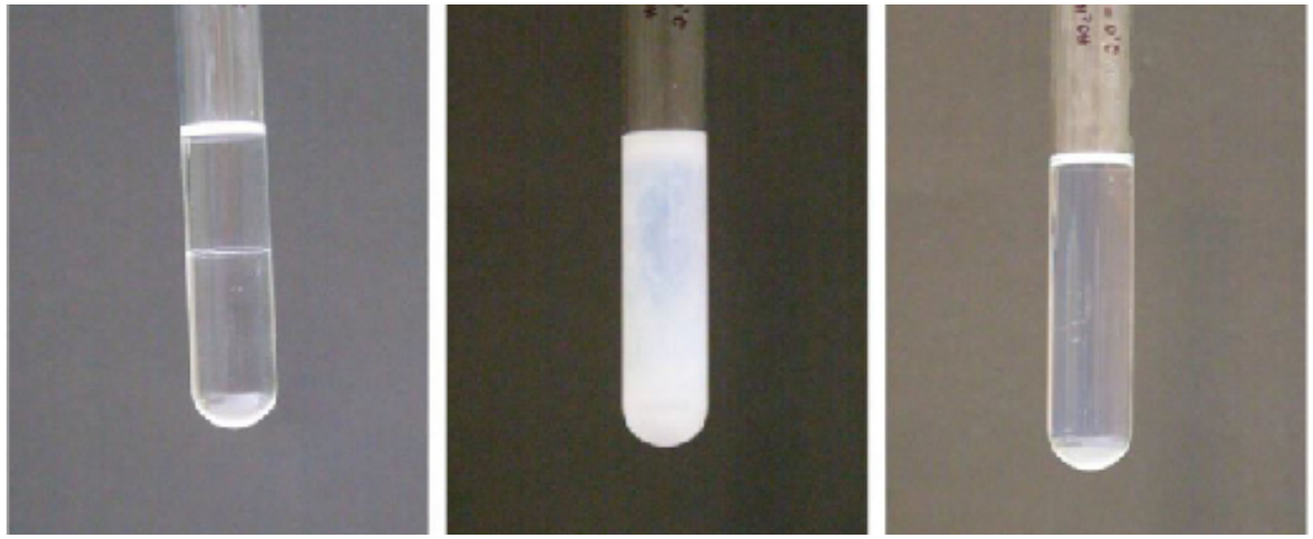
Outline

- What & why
- The model
- Dynamics of one particle
- Dynamics of two particles
- Monte Carlo simulation
- Conclusions

Phases of matter and critical points

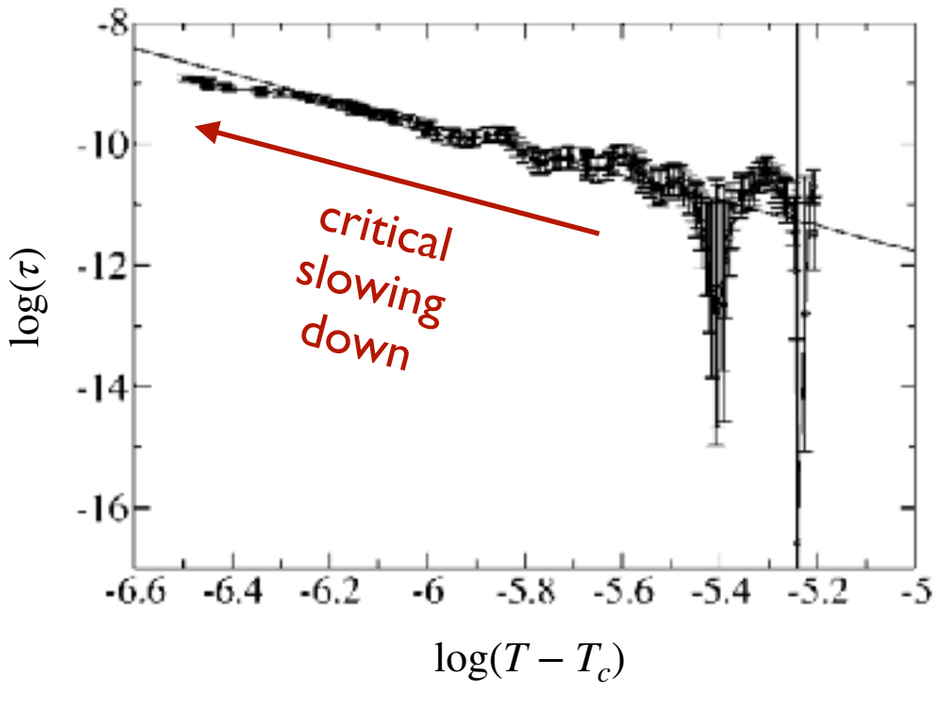




$T < T_c$ $T = T_c$ $T > T_c$

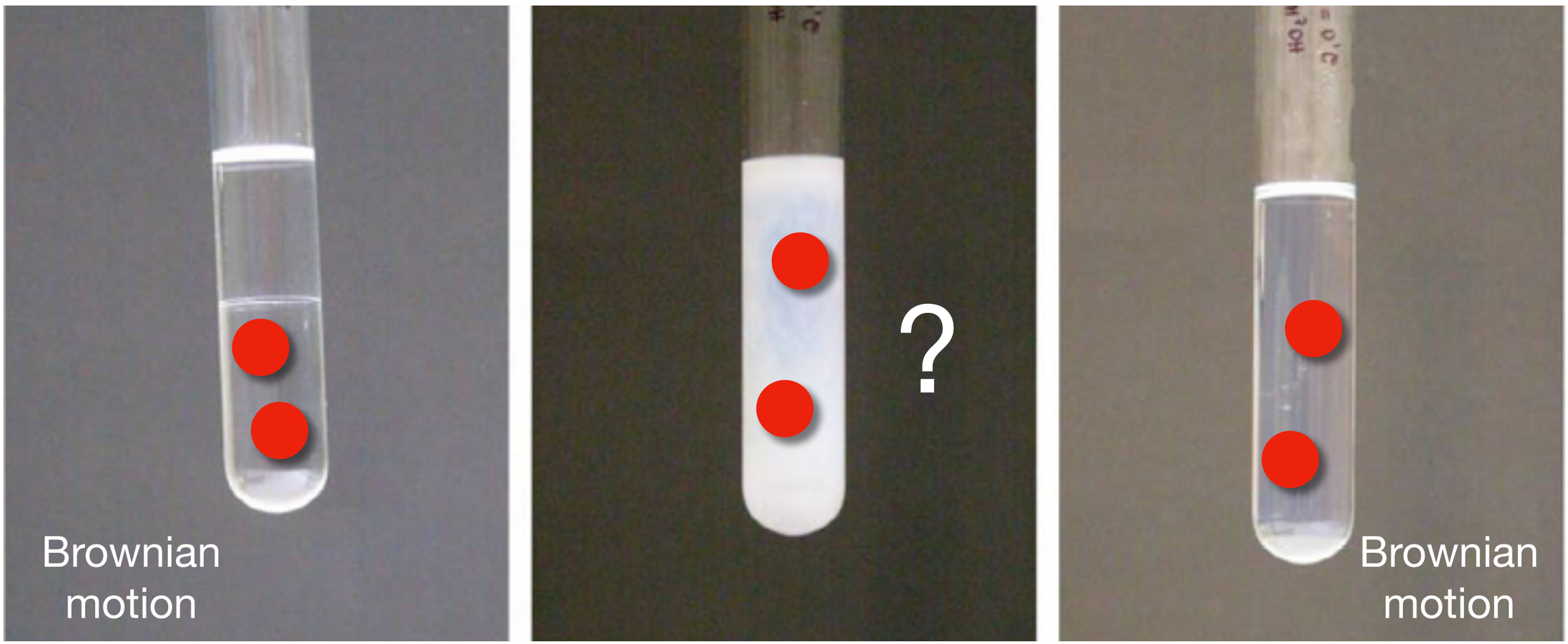
characteristic length: $\xi \sim (T - T_c)^{-\nu}$

characteristic time: $\tau \sim \xi^z$



Phases of matter and critical points

What happens to the dynamics of mesoscopic particles in a critical fluid?

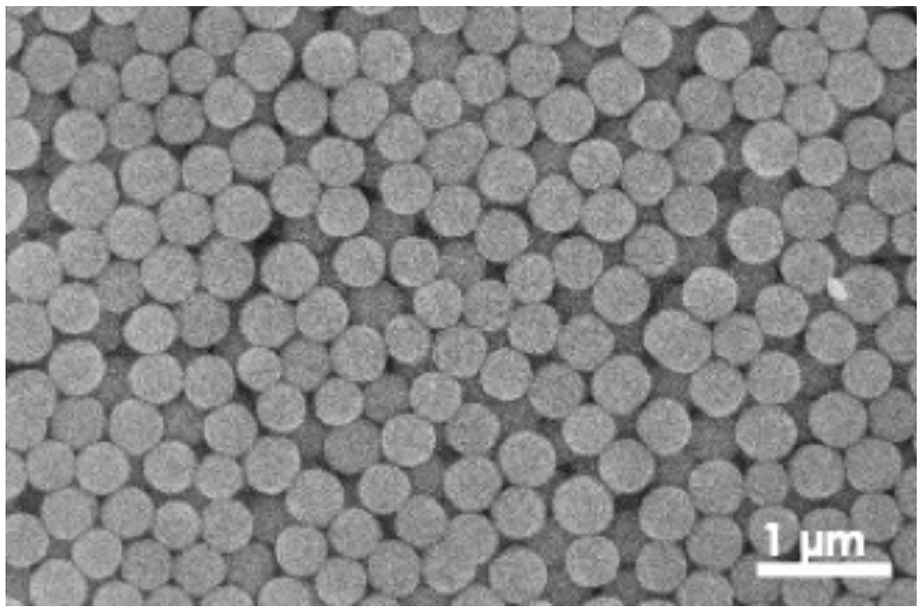


$T < T_c$ $T = T_c$ $T > T_c$

Why? Colloids in critical fluids

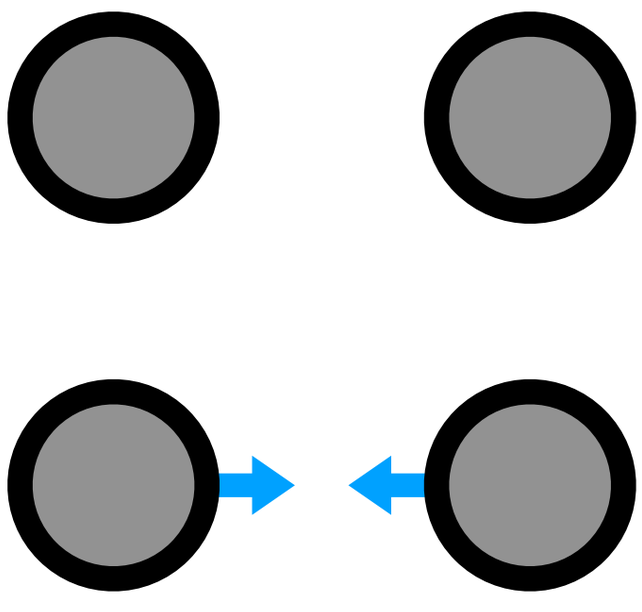
- Why colloids?
  - ⇨ testbed of statistical physics
  - ⇨ biological systems
  - ⇨ microrheology

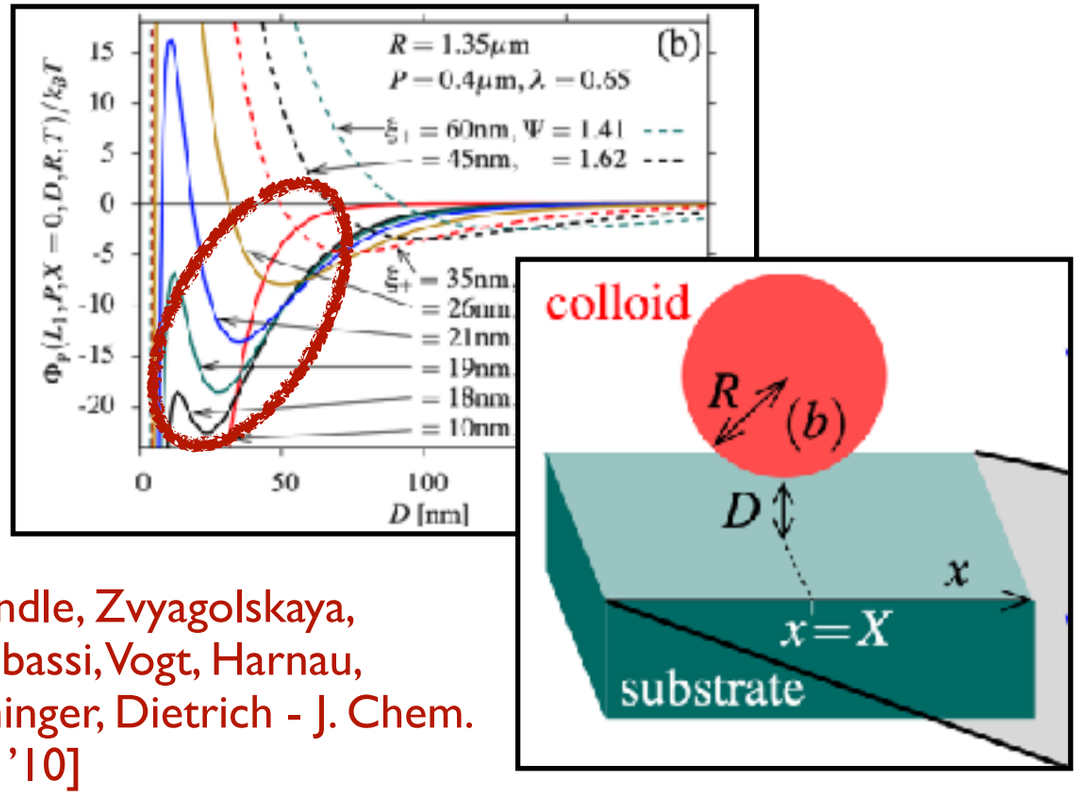
- Why colloids in critical fluids?
  - ⇨ critical Casimir force



$T \neq T_C$

$T = T_C$





[Tröndle, Zvyagolskaya, Gambassi, Vogt, Harnau, Bechinger, Dietrich - J. Chem. Phys '10]

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Modeling of a colloid

- Motion in a thermal bath

$$\dot{X} = -\nu \frac{\partial H}{\partial X} + \xi(t)$$

- In a harmonic trap $H = kx^2/2$

$$\tau_X = \frac{1}{\nu k}$$

$$\langle X(t) \rangle = X(0) \ \exp\left(-\frac{t}{\tau_X}\right)$$

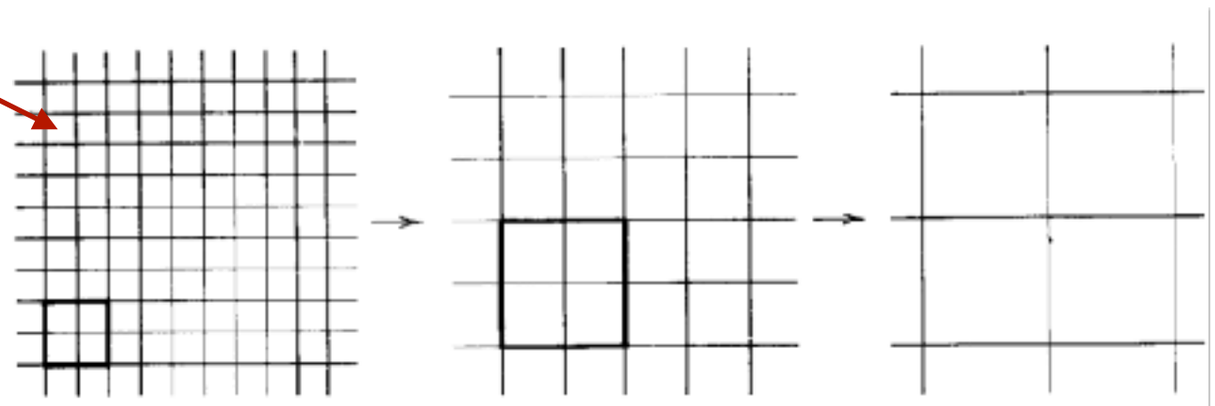
$$\langle X(t) X(t') \rangle = \frac{T}{k} \exp\left(-\frac{|t - t'|}{\tau_X}\right)$$

$$\ldots$$

Modeling of critical media

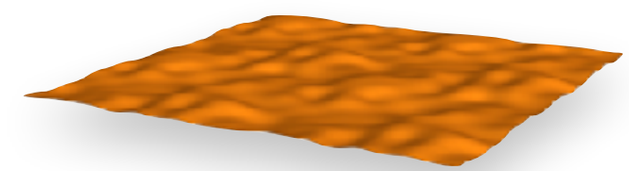
- Exploit scale invariance near critical point
  - ⇨ coarse-grain
  - ⇨ obtain field theory of order parameter

ϕ_i



$\cdots$

$\phi(x)$



$$H[\{\phi_i\}] = \sum_{\langle ij \rangle} \phi_i \phi_j + \ldots$$

$$H[\phi] = \int d^d x \left[\frac{1}{2}(\nabla \phi)^2 + \frac{1}{2} r \phi^2 + \frac{1}{4!} u \phi^4 + \ldots \right]$$

- ⇨ $\phi(x)$ evolves slower than microscopic d.o.f.
- ⇨ any* dynamics collapses to

$$\partial_t \phi(x) = -D \frac{\delta H}{\delta \phi(x)} + \zeta(x,t)$$

stochastic forcing

relaxation to eq.

Modeling of critical media

- …any dynamics up to conservation laws
- No conservation laws

Model A

$$\partial_t \phi(x) = -D \frac{\delta H}{\delta \phi(x)} + \zeta(x,t)$$

- Conserved field

Model B

$$\partial_t \phi(x) = D \nabla^2 \frac{\delta H}{\delta \phi(x)} + \zeta(x,t)$$

- OK for e.g. ferromagnets
  Not OK for real fluids (model H)

Model A and B in Gaussian approximation

$$H[\phi] = \int d^d x \left[\frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} r \phi^2 \right]$$

$$r = \xi^{-2} \propto T - T_C$$

$r > 0$: non-critical

$r = 0$: critical

- Dynamics diagonal in Fourier space
- Model A: each mode ϕ_q evolves with timescale

$$\tau_\phi = \frac{1}{D(r + q^2)}$$

off-criticality slowest mode is $\tau_\phi = 1/Dr$

at criticality $\tau_\phi \sim q^{-2} \to \infty$

dyn. crit. exp z

- Model B

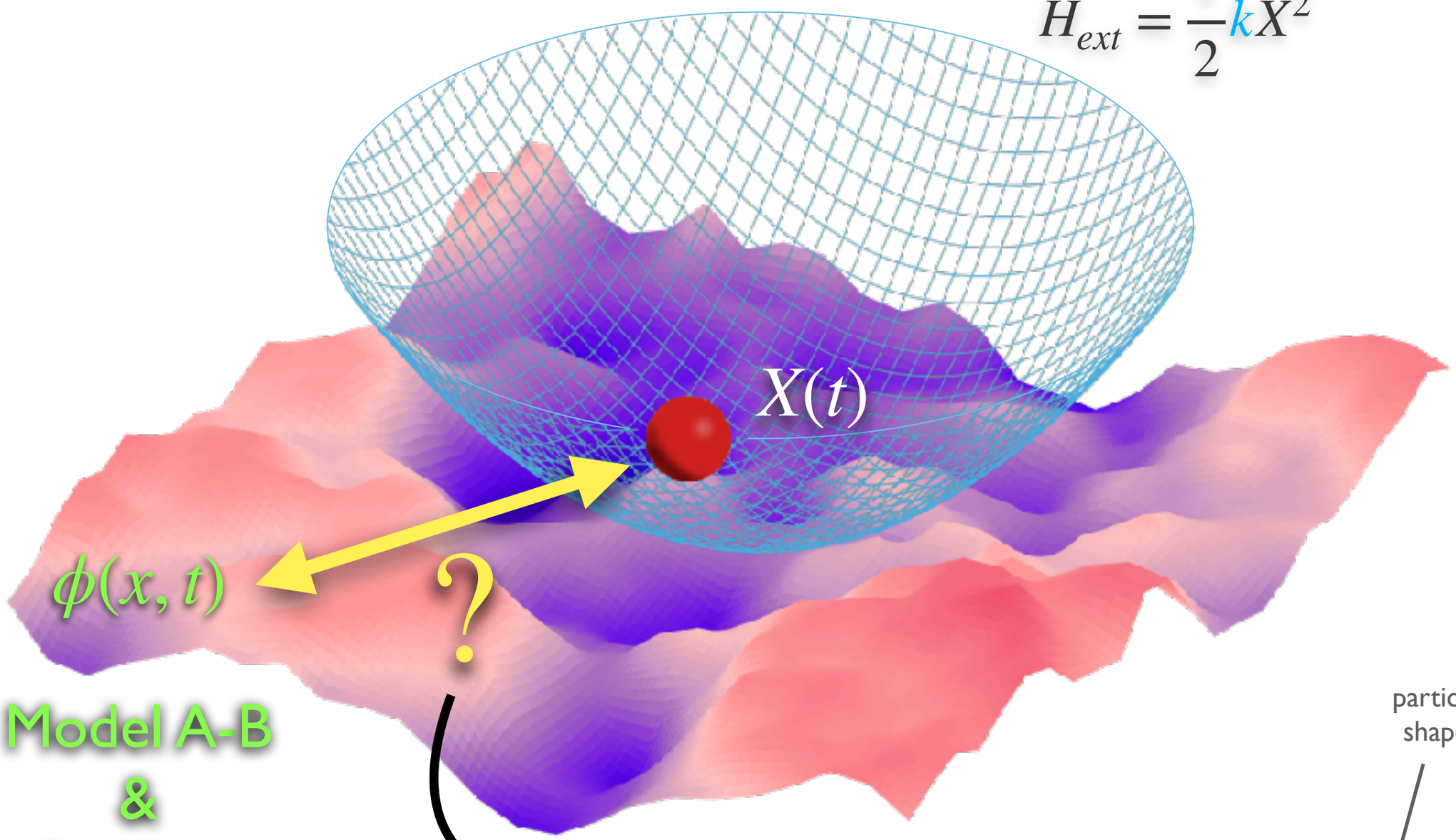
$$\tau_\phi = \frac{1}{Dq^2(r + q^2)}$$

off-criticality $\tau_\phi \sim q^{-2} \to \infty$

at criticality $\tau_\phi \sim q^{-4} \to \infty$

The model: which interaction?

$$H_{ext} = \frac{1}{2} k X^2$$

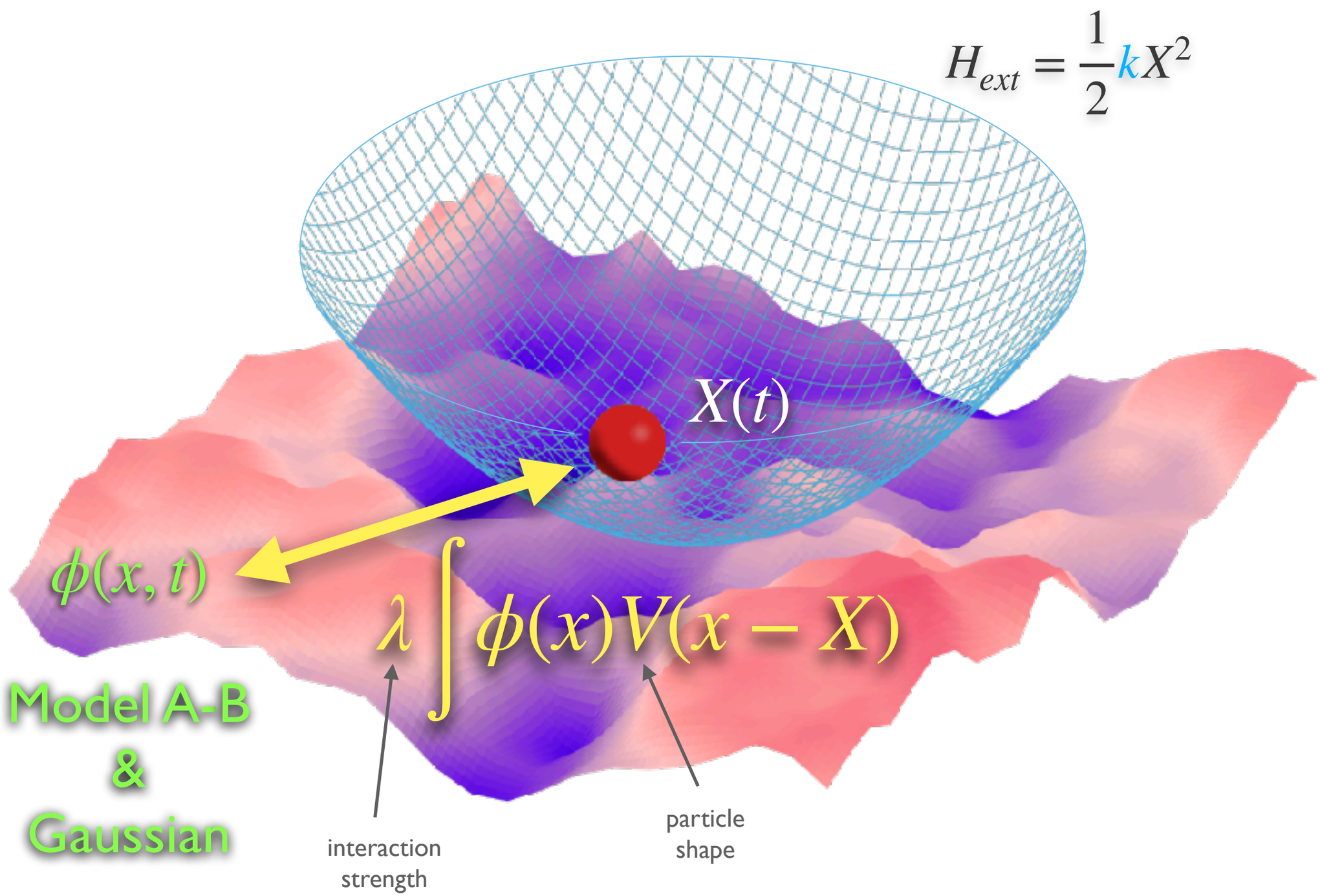


Model A-B
&
Gaussian

particle shape

$$\int dx \left[\phi(x) + \phi(x)^2 + \ldots + \nabla \phi(x) + \ldots \right] V(x - X)$$

The model: which interaction?



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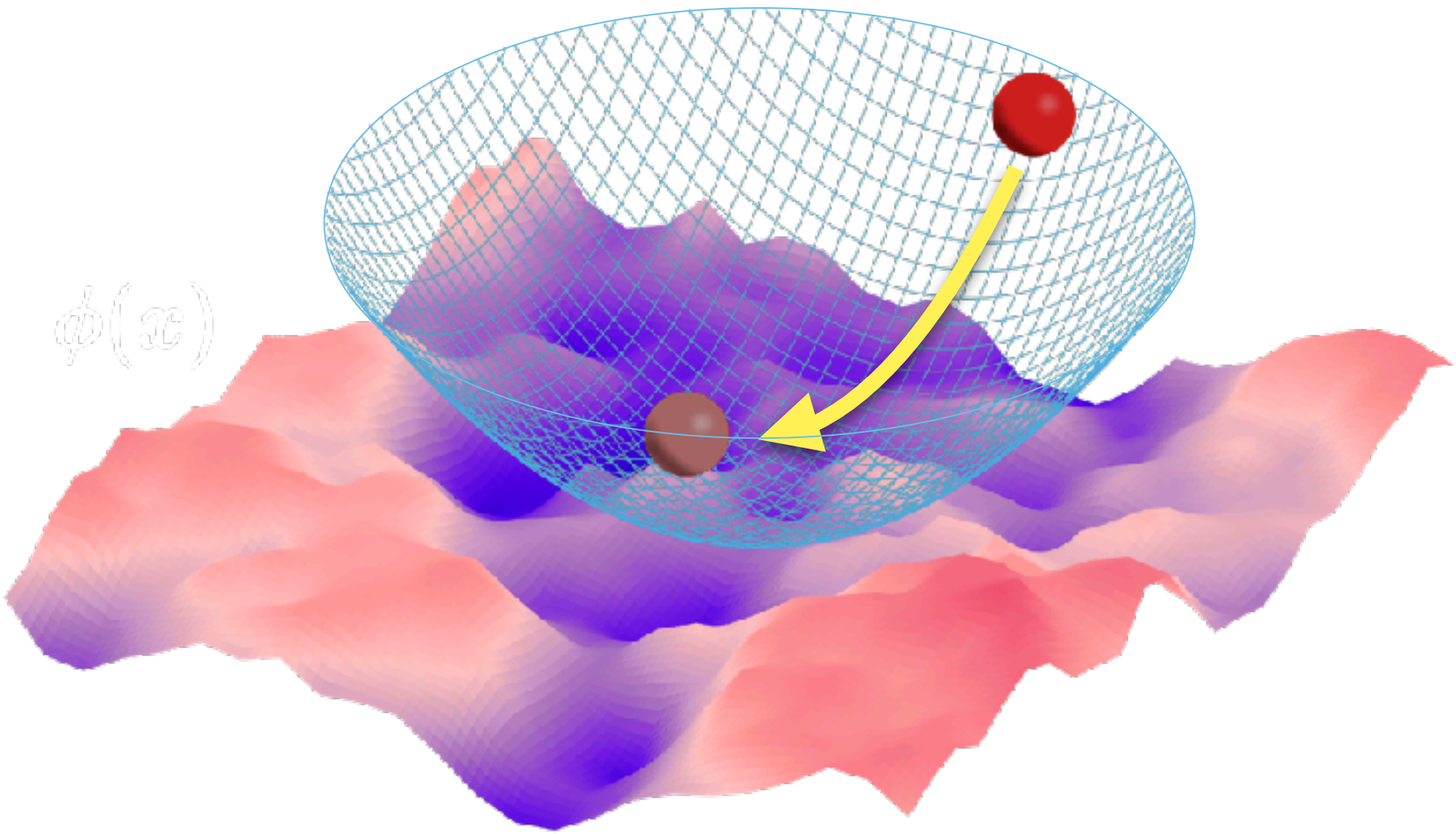
Relaxation to equilibrium: model A

$$\langle X(t) \rangle$$

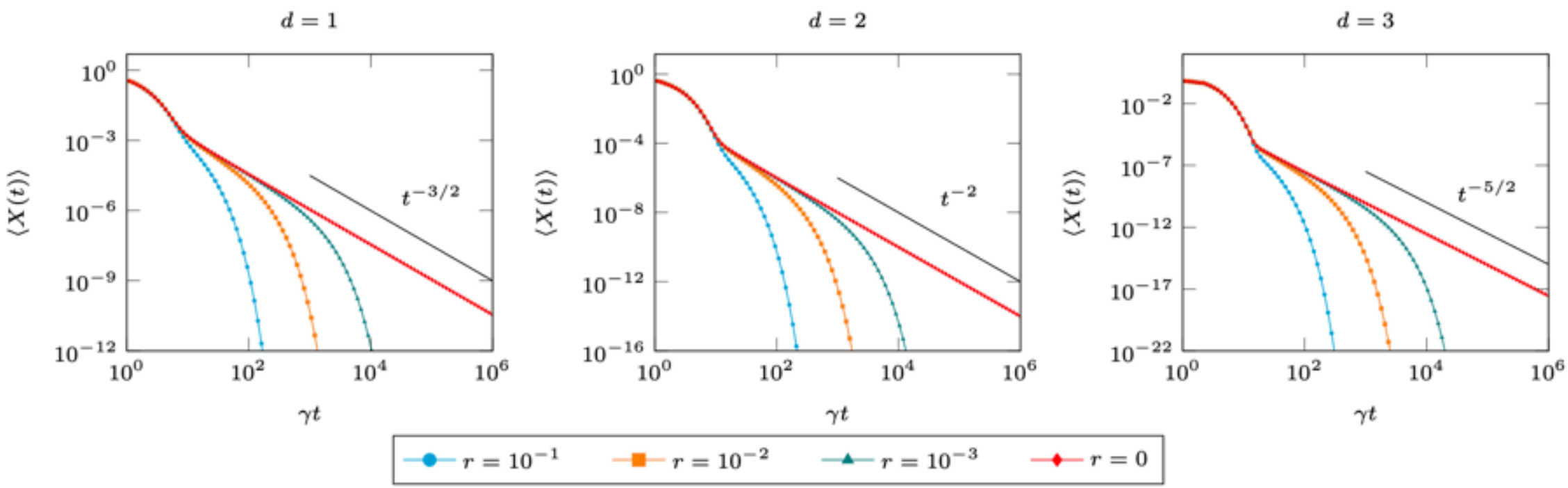


Relaxation to equilibrium: model A

$$\langle X(t)\rangle \sim \lambda^2 \begin{cases} e^{-t/\tau_X} & \tau_X > \tau_\phi \\ e^{-t/\tau_\phi} & \tau_\phi > \tau_X \\ t^{-(d/2+1)} & r = 0 \end{cases} + o\left(\lambda^4\right)$$

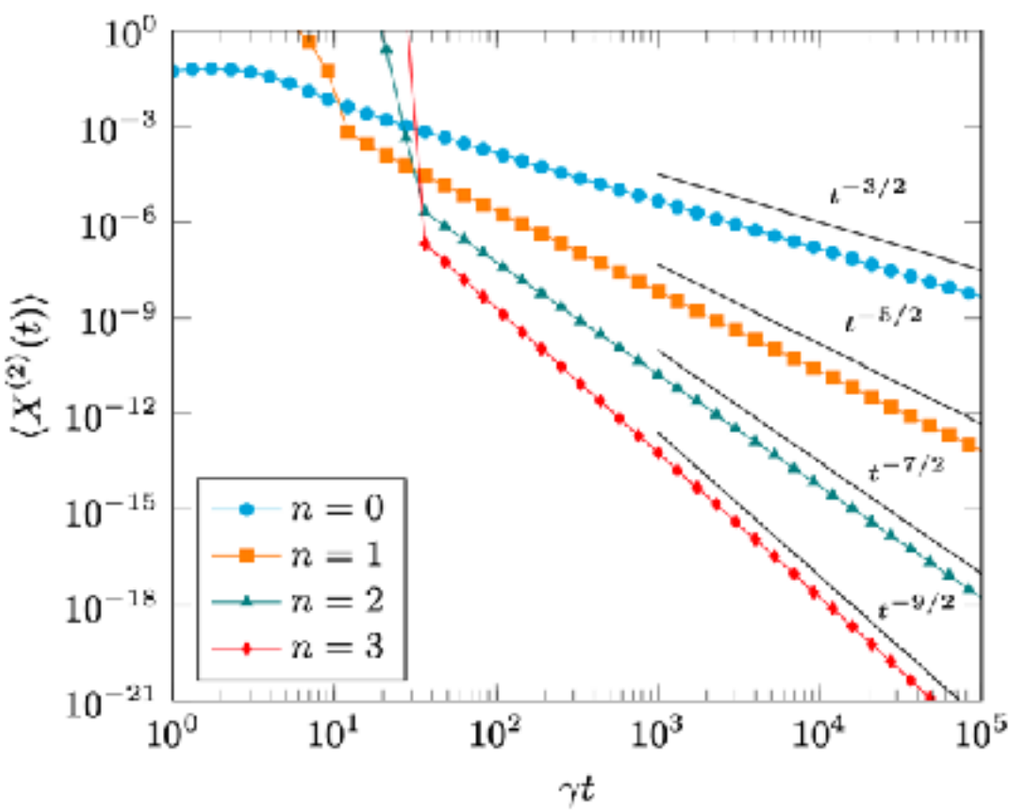
with:

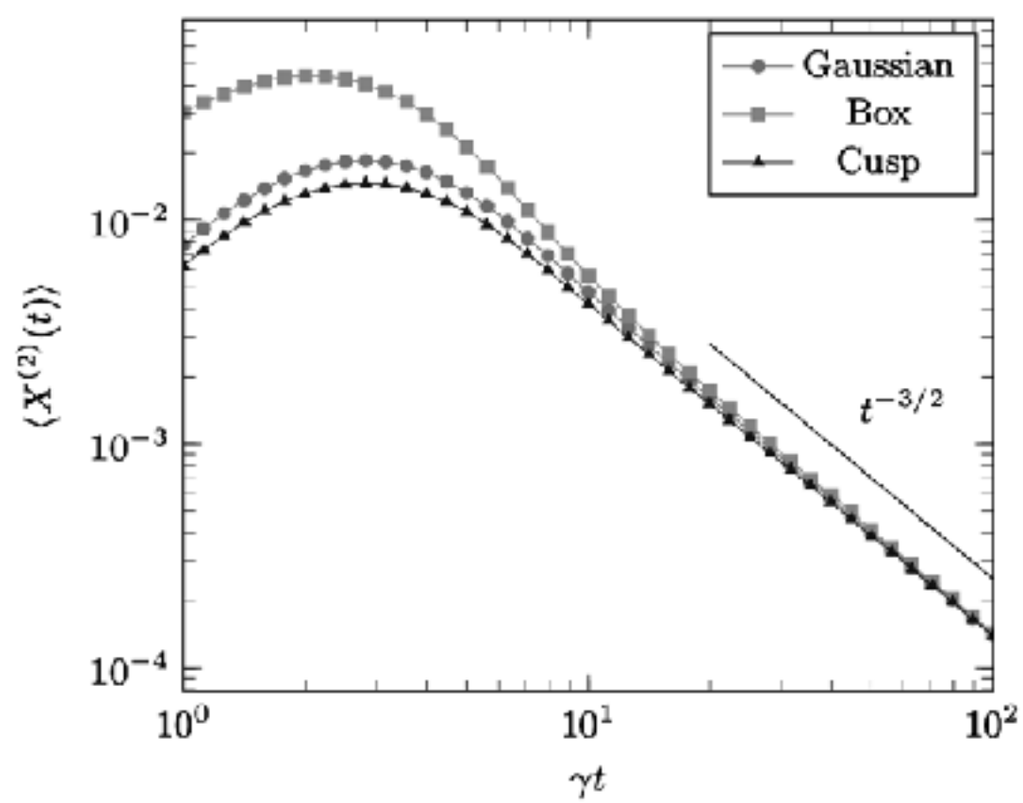
$$\tau_x = 1/\nu k$$
$$\tau_\phi = 1/Dr$$



Universality of long-time behaviour

$$
\begin{aligned}
H_{int} &= \int d^d x\, \phi(x) V(x - X) \\
&= \int \frac{d^d q}{(2\pi)^d} \phi_q e^{iq\cdot X} V_q \\
&= \int \frac{d^d q}{(2\pi)^d} \phi_q e^{iq\cdot X} \left(c_0 + c_1 q + c_2 q^2 + c_3 q^3 + \ldots \right)
\end{aligned}
$$





Autocorrelation: model A

- Same behaviour as relaxation

$$\langle X(t) \cdot X(0) \rangle \quad \sim \quad \lambda^2 \begin{cases} e^{-t/\tau_X} & \tau_X > \tau_\phi \\ e^{-t/\tau_\phi} & \tau_\phi > \tau_X \\ t^{-d/2} & r = 0 \end{cases}$$

- Also shown to be universal

Dynamics in model B conserved field

$$\langle X(t) \rangle \quad \sim \quad \lambda^2 \begin{cases} t^{-(d/2+2)} & r > 0 \\ t^{-(d/4+1)} & r = 0 \end{cases}$$

$$\langle X(t) \cdot X(0) \rangle \quad \sim \quad \lambda^2 \begin{cases} t^{-(d/2+1)} & r > 0 \\ t^{-d/4} & r = 0 \end{cases}$$

Critical behaviour of dynamical quantities

Model A

$$\langle X(t)\rangle \sim \lambda^2 \begin{cases} e^{-t/\tau_x} & \tau_X > \tau_\phi \\ e^{-t/\tau_\phi} & \tau_\phi > \tau_X \\ t^{-(d/2+1)} & r = 0 \end{cases}$$

$$\langle X(t)\cdot X(0)\rangle \sim \lambda^2 \begin{cases} e^{-t/\tau_X} & \tau_X > \tau_\phi \\ e^{-t/\tau_\phi} & \tau_\phi > \tau_X \\ t^{-d/2} & r = 0 \end{cases}$$

Model B

$$\langle X(t)\rangle \sim \lambda^2 \begin{cases} t^{-(d/2+2)} & r > 0 \\ t^{-(d/4+1)} & r = 0 \end{cases}$$

$$\langle X(t)\cdot X(0)\rangle \sim \lambda^2 \begin{cases} t^{-(d/2+1)} & r > 0 \\ t^{-d/4} & r = 0 \end{cases}$$

- Relaxation and correlation behaviour is related
- Relation between particle crit exp and field crit exp?

Non-perturbative fluctuation dissipation relation

- If system is slightly perturbed from equilibrium

$$\dot{X} = \ldots + f(t)$$

then response of an observable

$$\langle O(t) \rangle = \int dt' R(t - t') f(t')$$

is related to its equilibrium correlations as

$$R(t) \propto \frac{d}{dt} \langle O(t) O(0) \rangle_{eq}$$

- Setting $X(0) = X_0$ is equivalent to $f(t) = X_0 \delta(t)$

- Taking $O(t) = X(t)$

$$\langle X(t) \rangle \propto \frac{d}{dt} \langle X(t) X(0) \rangle_{eq} \quad \longrightarrow \quad \langle X(t) \rangle \propto t^{-1} \langle X(t) X(0) \rangle_{eq}$$

Relation between field and particle exponents (tentative)

- Take perturbative correction e.g.

$$\left\langle X^{(2)}(t)\right\rangle = \nu \int \frac{d^d q}{(2\pi)^d} iq\,|V_q|^2 \int_{t_0}^{t} ds_2 e^{-\gamma(t-s_2)} \int_{t_0}^{s_2} ds_1 \times \left[\chi_q\left(s_1,s_2\right) + \nu q^2 e^{-\gamma(s_2-s_1)} C_q\left(s_1,s_2\right)\right] Q_q\left(s_1,s_2\right)$$

field corr. func.

particle "caracteristic" function

field susceptibility

- Plug in scaling form of field functions

$$\langle X(t)\rangle \sim t^{-(d/z+1)}$$

- This only applies for Gaussian theories…

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Two particles

- Model A

$$\langle X_1(t)X_2(0)\rangle \sim \lambda^2 \begin{cases} e^{-t/\tau_X} & \tau_X > \tau_\phi \\ e^{-t/\tau_\phi} & \tau_\phi > \tau_X \\ t^{-d/2} & r = 0 \end{cases}$$

- Model B

$$\langle X_1(t)X_2(0)\rangle \sim \lambda^2 \begin{cases} t^{-(d/2+1)} & r > 0 \\ t^{-d/4} & r = 0 \end{cases}$$

- Same asymptotic of single-particle autocorrelation…

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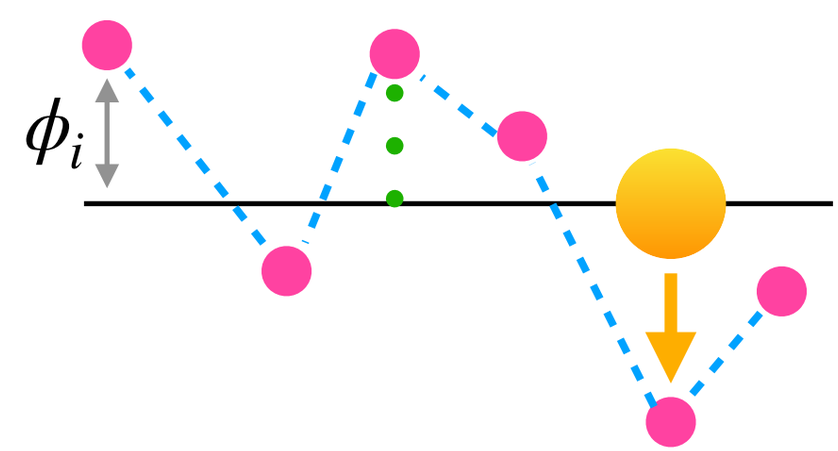
Simulation algorithm and lattice polymer mapping

- Field: discretized equation



$$\partial_t \phi_i = \frac{D}{\Delta x^2}\left(\phi_{i+1} + \phi_{i-1} - 2\phi_i\right) + \zeta_i \quad - Dr\phi_i \quad + \quad \frac{D\lambda}{\Delta x}\delta_{i,X_t}$$

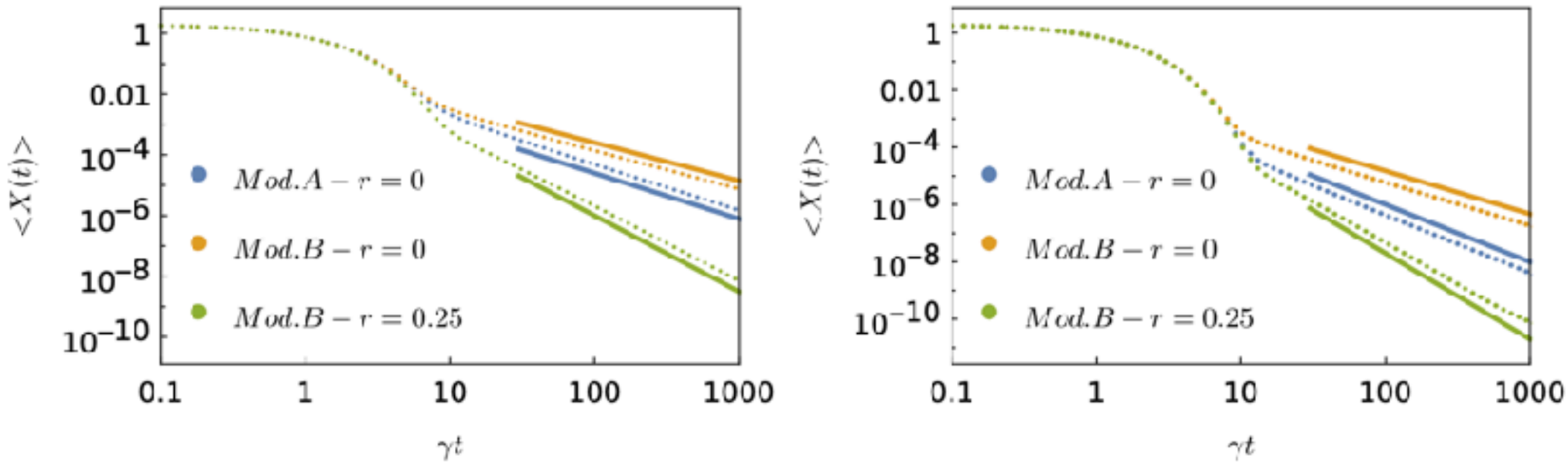
Rouse model

- Particle: random walker

$$W(i \to i \pm +1) \propto \min\left\{1, \exp\left[-\frac{H(i \pm 1) - H(i)}{T}\right]\right\}$$

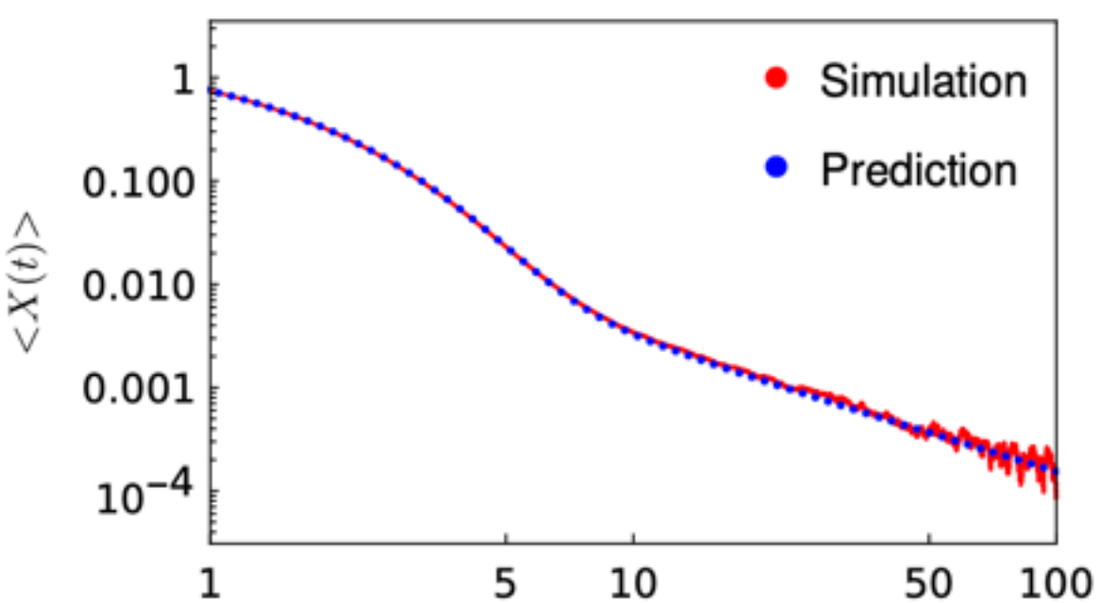
Simulation algorithm and lattice polymer mapping

Low T, $\lambda = 2$

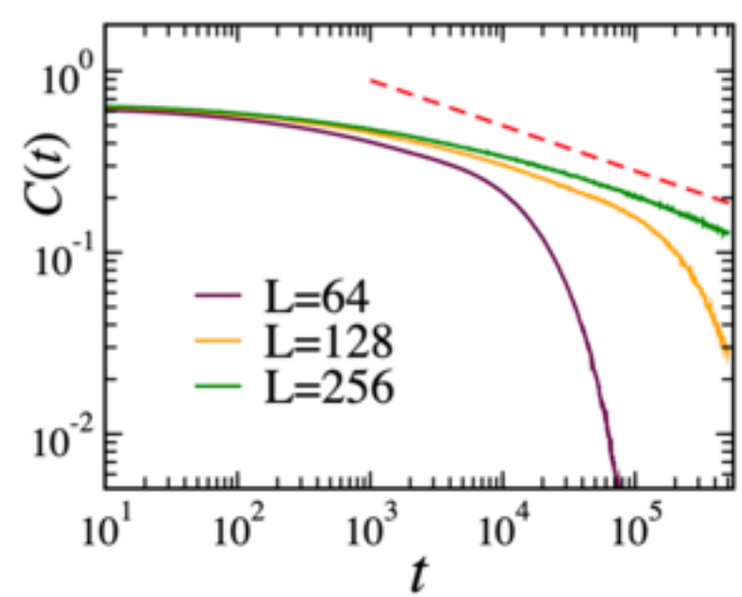


(a) $d = 1$ (b) $d = 2$

High T, $\lambda = 0.5$



Finite-size scaling



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Summary of results

- Equilibrium behaviour [see thesis]
- Critical exponents and universality of relaxation, autocorrelation, cross-correlation
- Connection to field scaling exponent z, e.g.

$$\langle X(t) \rangle \sim t^{-(d/z+1)}$$

- Fluctuation-dissipation relation

$$\langle X(t) \rangle \sim \frac{d}{dt} \langle X(t) X(0) \rangle$$

- Non-perturbative numerical validation

Perspectives

- Beyond perturbation theory
- Inclusion of ϕ^4 term

- $\phi^2, \phi^3, \ldots$ field-particle coupling
- Explore other universality classes

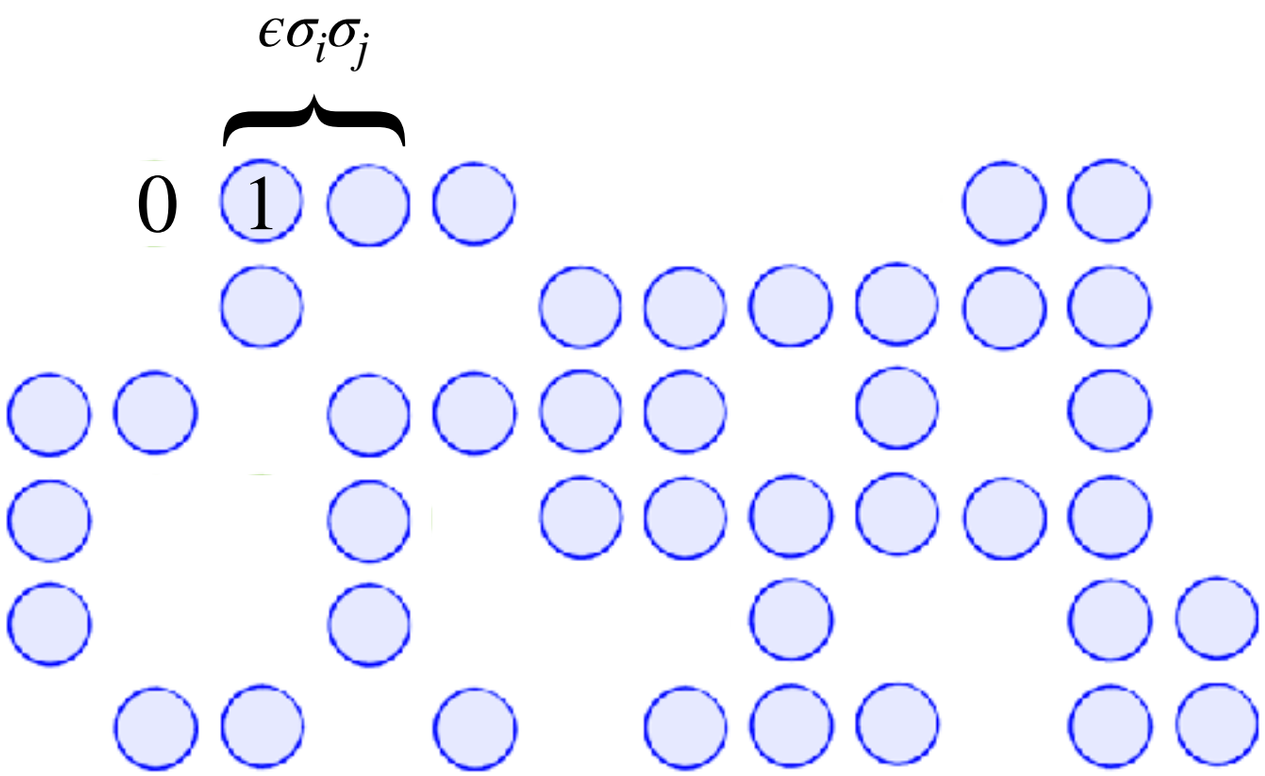
- Multiple and/or active particles

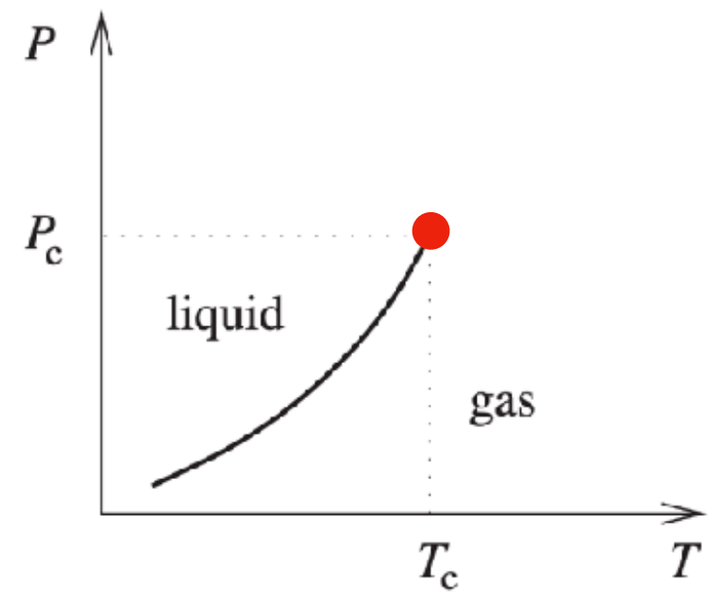
Thank you for your attention!

Extra slides

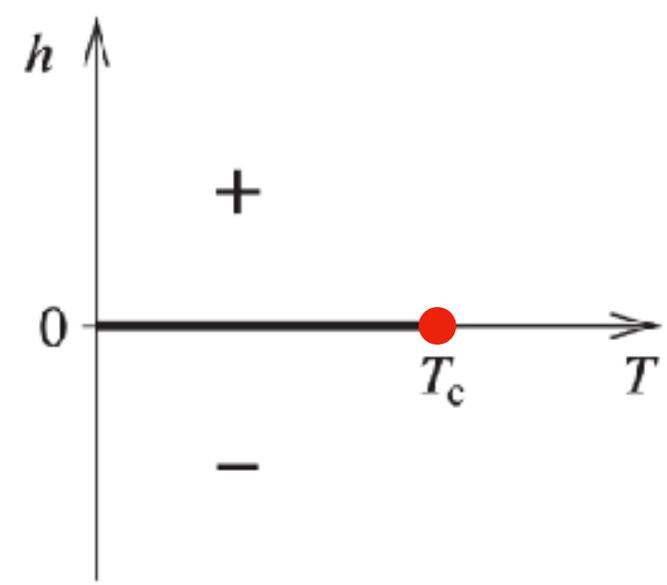
Extra

Mapping of single-component fluid to Ising



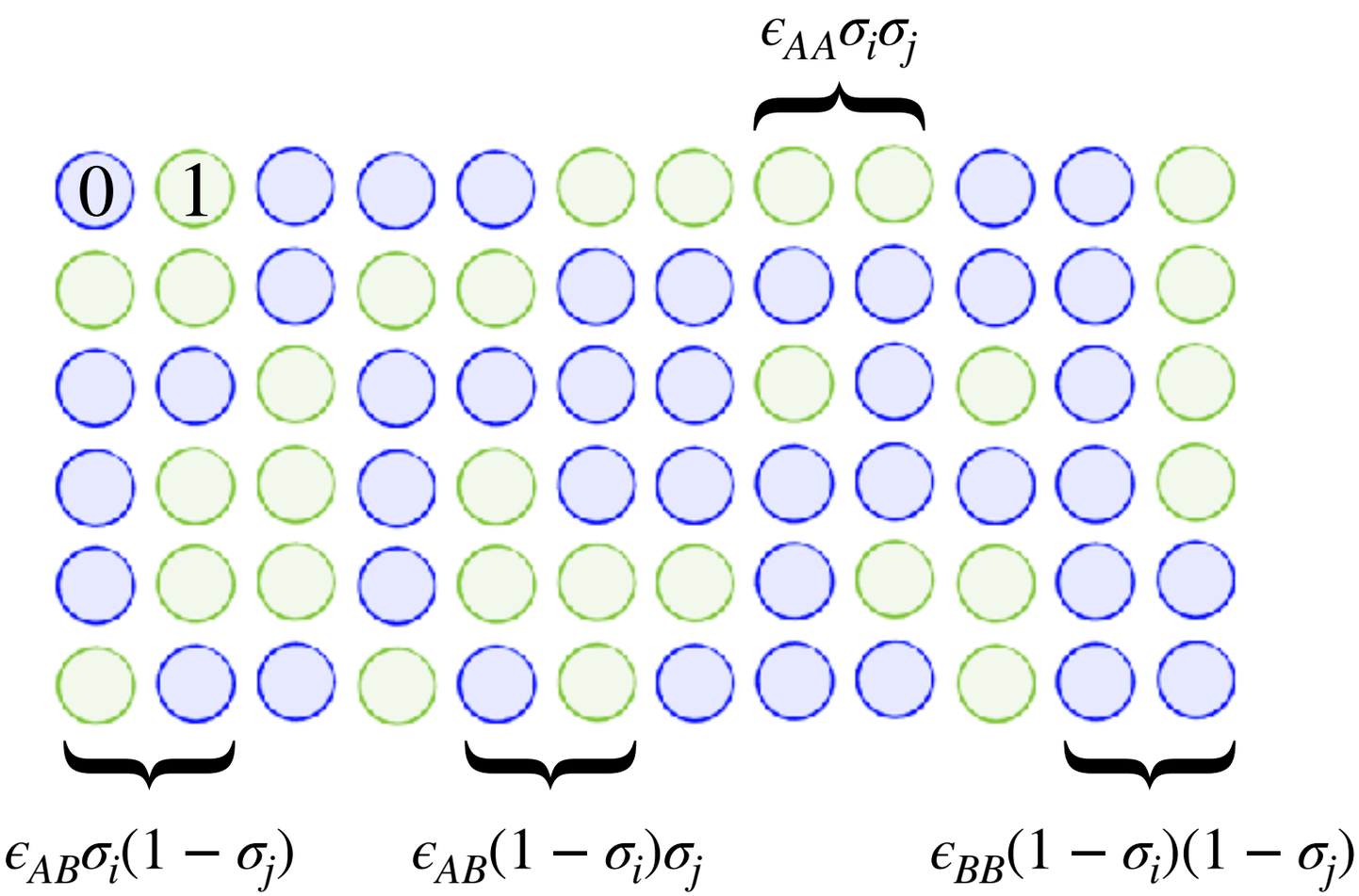


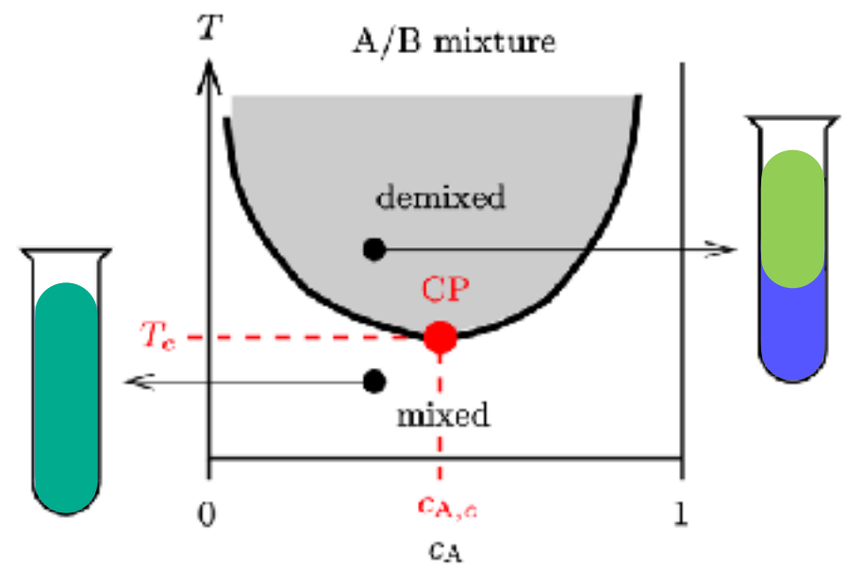
$$H[\{\sigma\}] = \sum_{\langle i,j \rangle} \epsilon\sigma_i\sigma_j$$



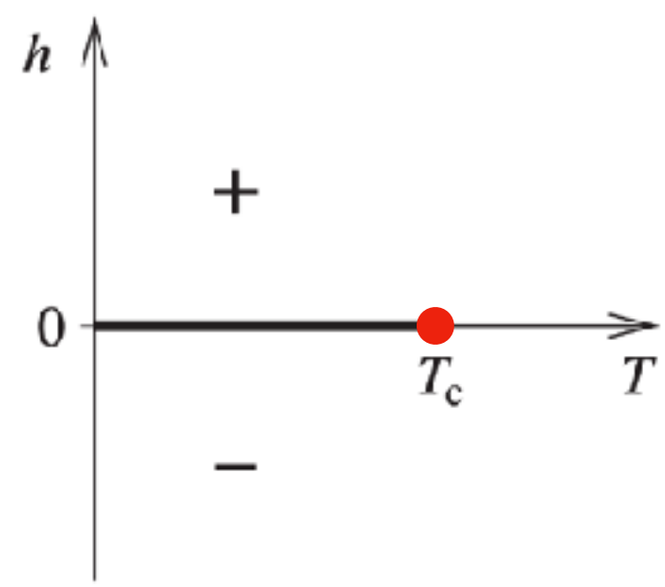
Extra

Mapping of binary mixture to Ising

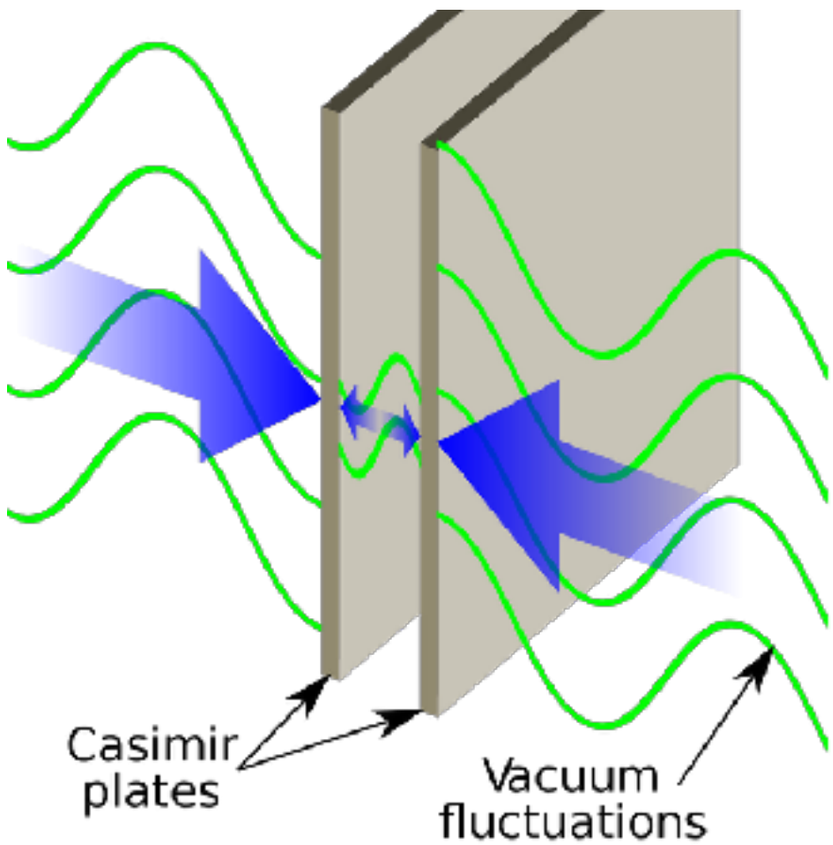




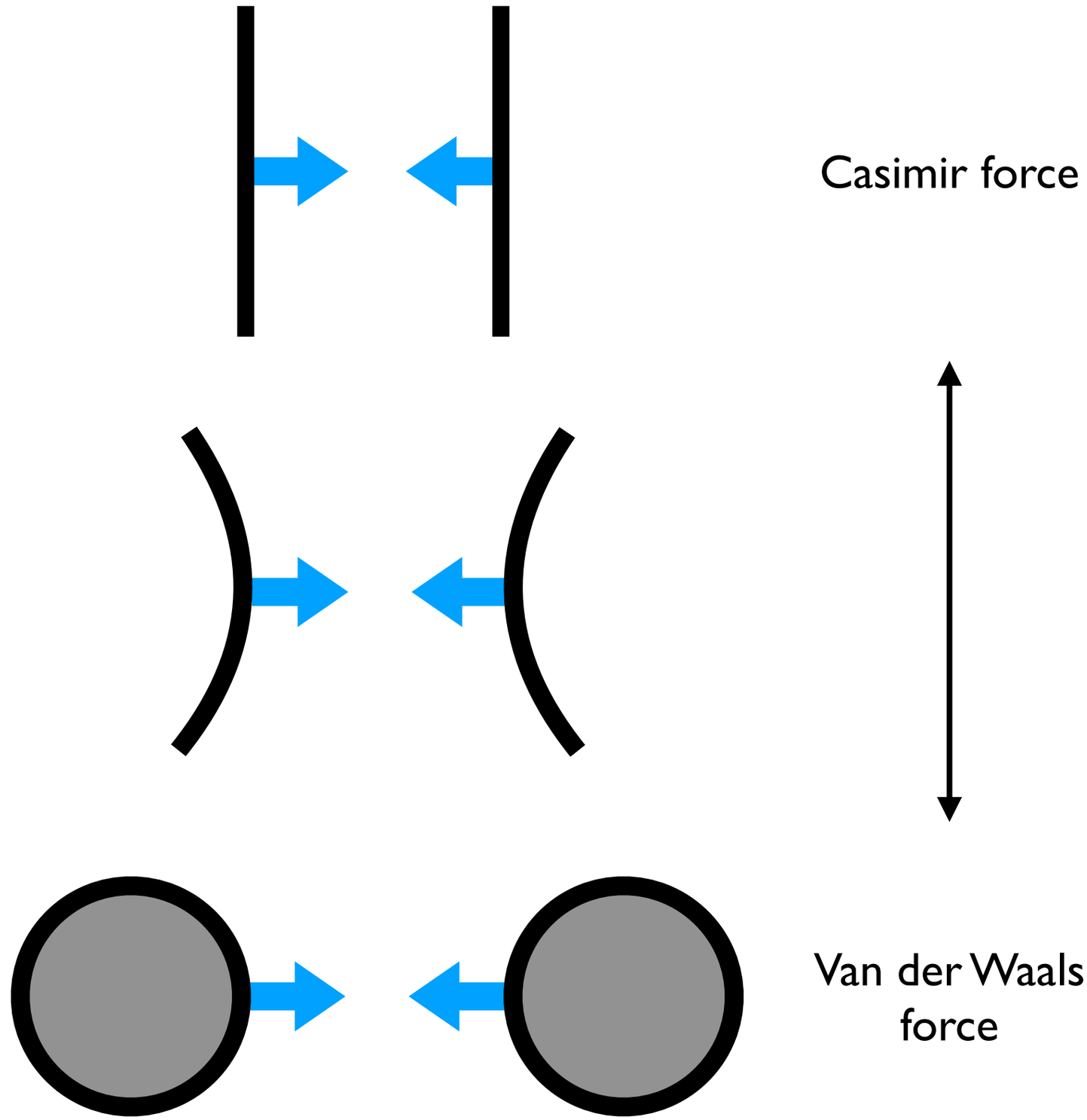
$$H[\{\sigma\}] = \sum_{\langle i,j\rangle}\left[\epsilon_{AA}\sigma_i\sigma_j + \epsilon_{AB}\sigma_i(1-\sigma_j) + \ldots\right]$$
$$= A\sum_{\langle i,j\rangle}\sigma_i\sigma_j + B\sum_i \sigma_i$$



Why colloids in critical fluids? Casimir force in QED



$$E \propto -\frac{\hbar c}{L^3}$$



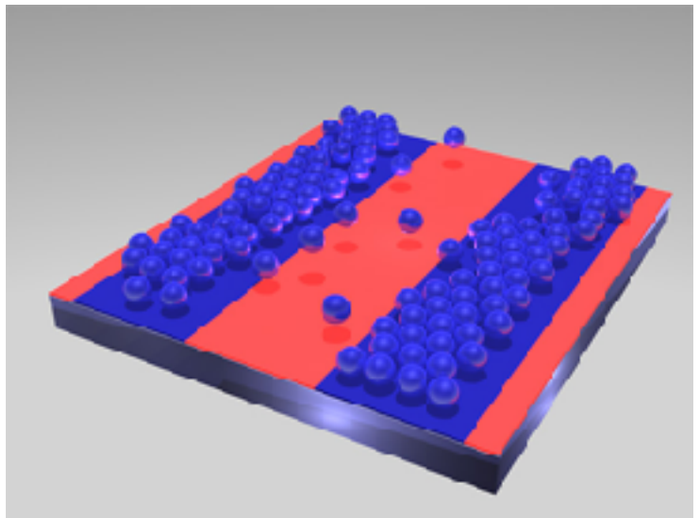
Casimir force in QED and StatPhys

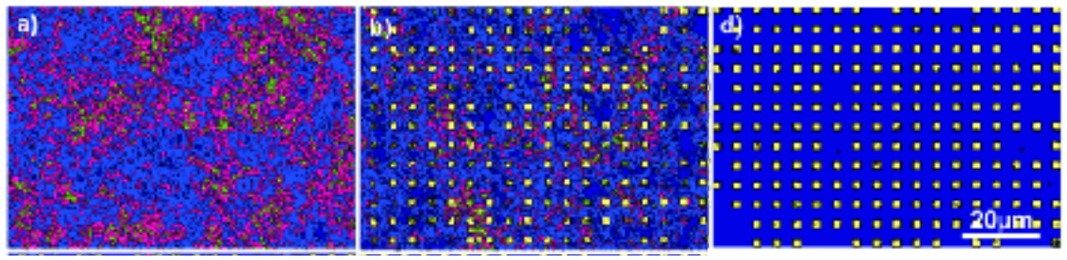
	QED	**StatPhys**
Fluctuating quantity	E, B	Order parameter ϕ
Source of fluctuations	Quantum $\hbar > 0$	Thermal $kT > 0$
Range of fluctuations	∞ $(m_\gamma = 0)$	Correlation length ξ ∞ at CP
Resulting force after confinement	Long-range	**Variable range ξ** Long range at CP

[Review: Gambassi - JPCS '09]

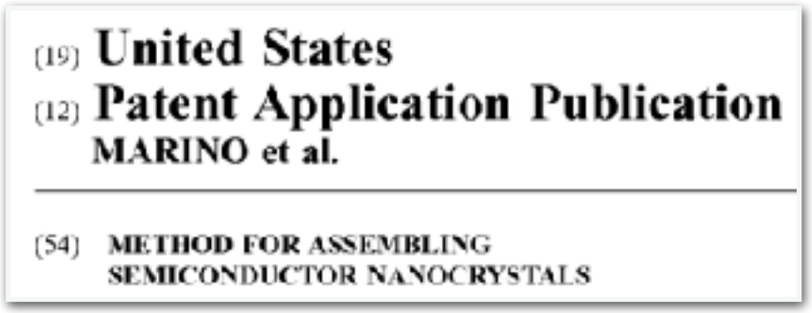
Critical Casimir force and patterned surfaces

Aggregation and assembly

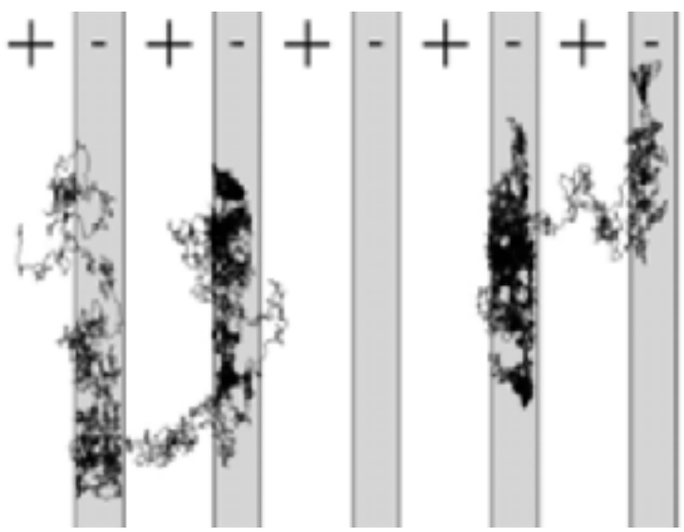




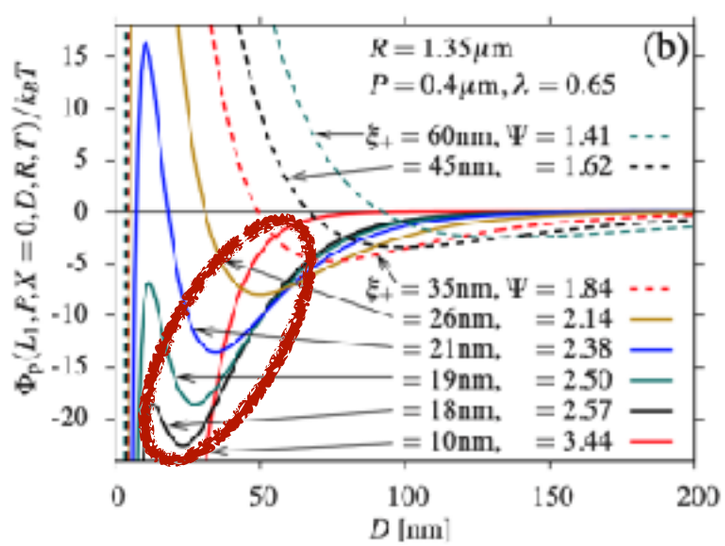
$T \longrightarrow T_c$

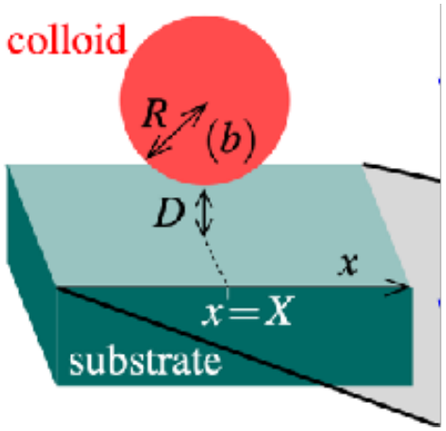

(19) United States
(12) Patent Application Publication
MARINO et al.

(54) METHOD FOR ASSEMBLING SEMICONDUCTOR NANOCRYSTALS



Levitation





[Review: Gambassi, Dietrich - Soft Matter '10]

Equilibrium distribution of the particle

$$P_{eq}(X) \propto \int [d\phi] e^{-\beta H[\phi,X]}$$

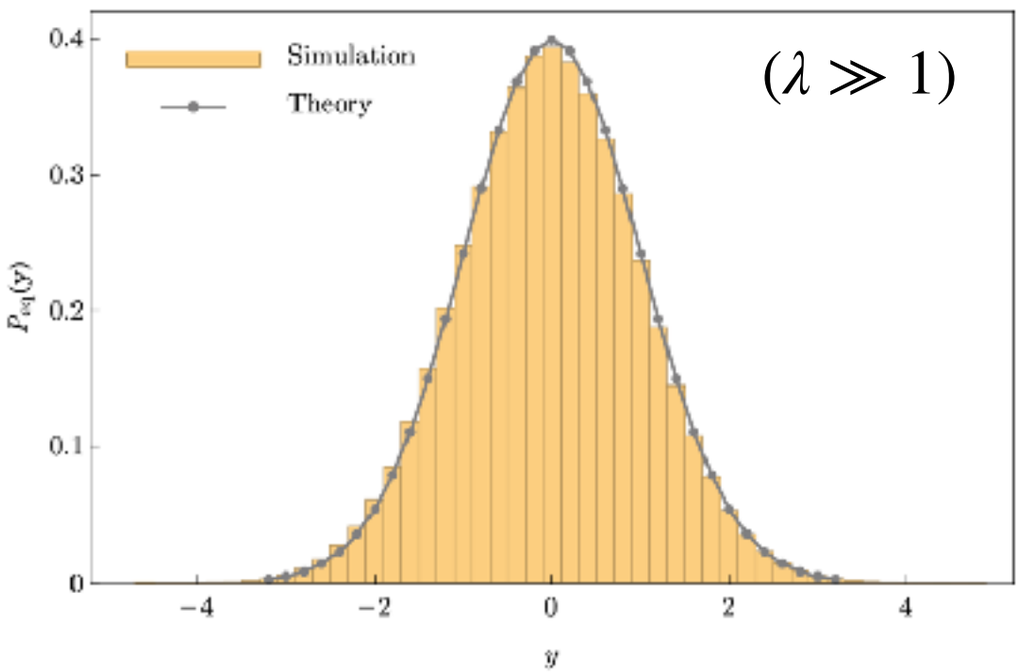
$$= e^{-\beta H_X} \cdot \int [d\phi] e^{-\beta(H_\phi + H_{int})}$$

$$\propto e^{-\beta H_X}$$

$$\psi(x) = \phi(x+X)$$

$$[d\phi] = [d\psi]$$

$$H_\phi[\phi] = H_\phi[\psi]$$

$$H_{int}[\phi, X] = H_{int}[\psi, 0]$$

Any field theory
Any field-particle coupling
Any external potential